Network & Distributed Programming

The Internet: Protocols and Layers

Lecture 2



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



DIETI

UNINA

A “nuts and bolts” view of the Internet



Billions of connected computing *devices*:

- *hosts = end systems*
- running *network apps* at Internet’s “edge”



Packet switches: forward packets (chunks of data)

- *routers*, *switches*

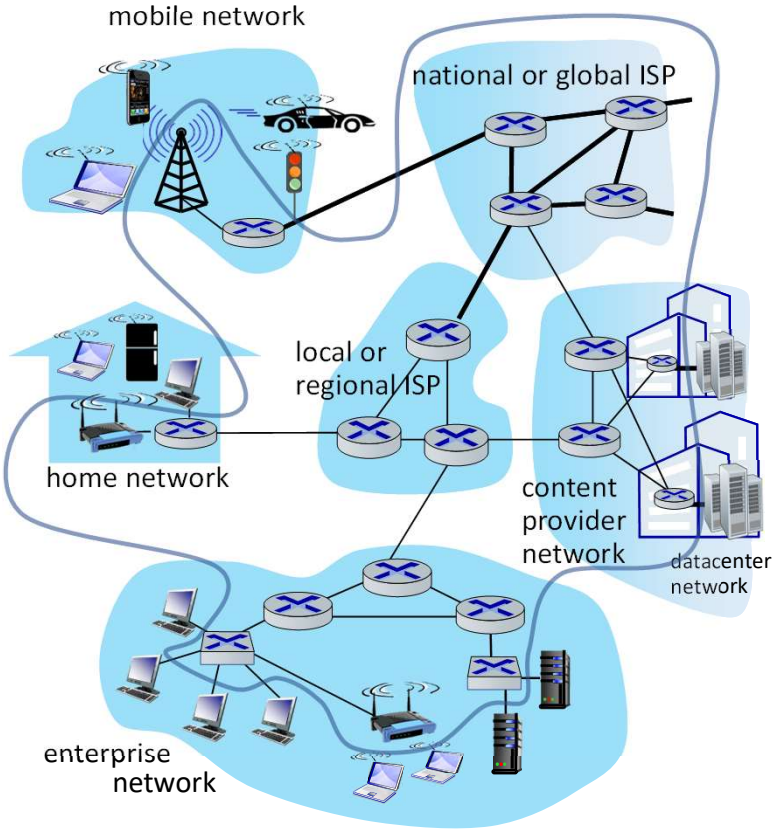


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



Various connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt: monitor energy use



Security Camera



Slingbox: remote control cable TV



Web-enabled toaster + weather forecaster



AR devices

Internet phones





sensorized, bed mattress

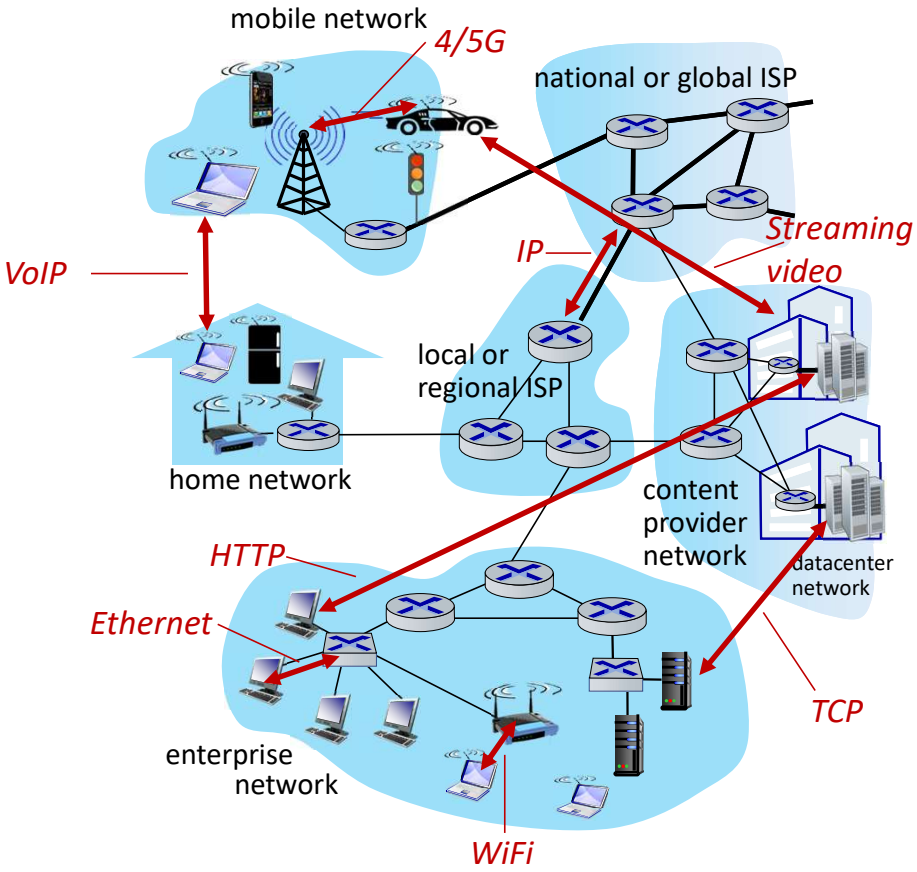


Fitbit

...and others

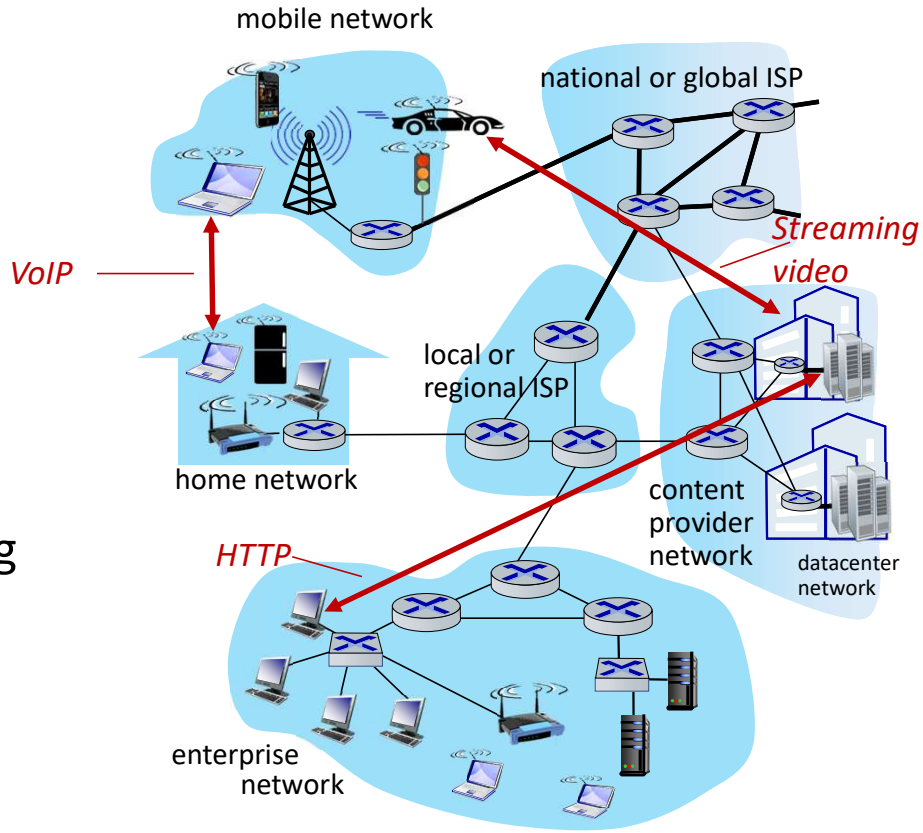
A “nuts and bolts” view of the Internet

- *Internet:* “network of networks”
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP, DNS, TCP, UDP, IP, WiFi and Ethernet protocol families, etc.
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



A services view of the Internet

- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …
- provides *programming interface* to distributed applications:
  - "interfaces" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service



Protocols

Human protocols:

- say “Hello” as a greeting
- respond to the greeting
- raise your hand to ask a question
- wait until you are allowed to speak
- Ask the question
- wait for the answer

… specific messages sent

… specific actions taken when message received, or other events

Network protocols:

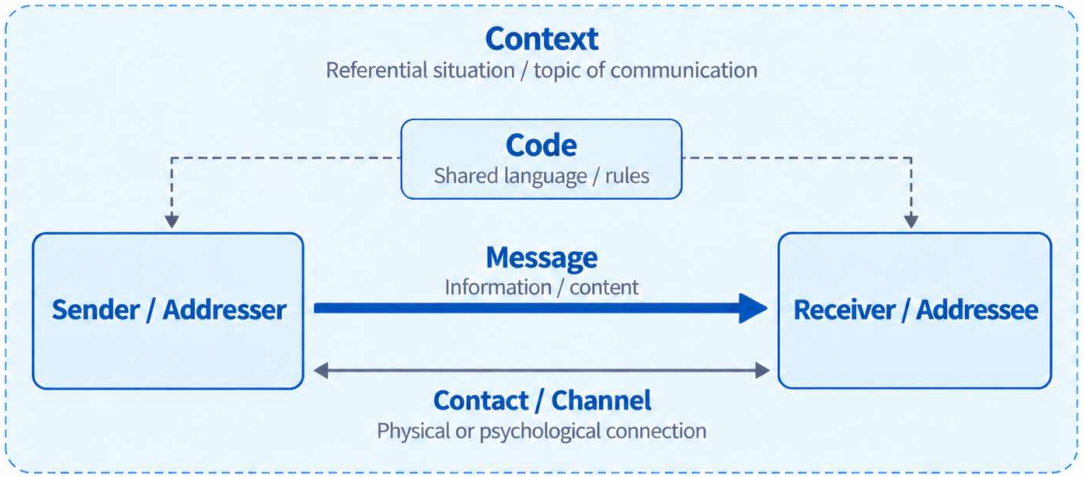
- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

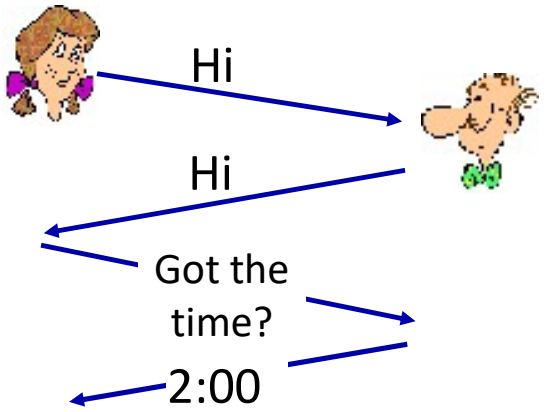
Protocols define the format, order of messages sent and received among network entities, and actions taken on msg transmission, receipt

Human and network protocols

A human protocol:

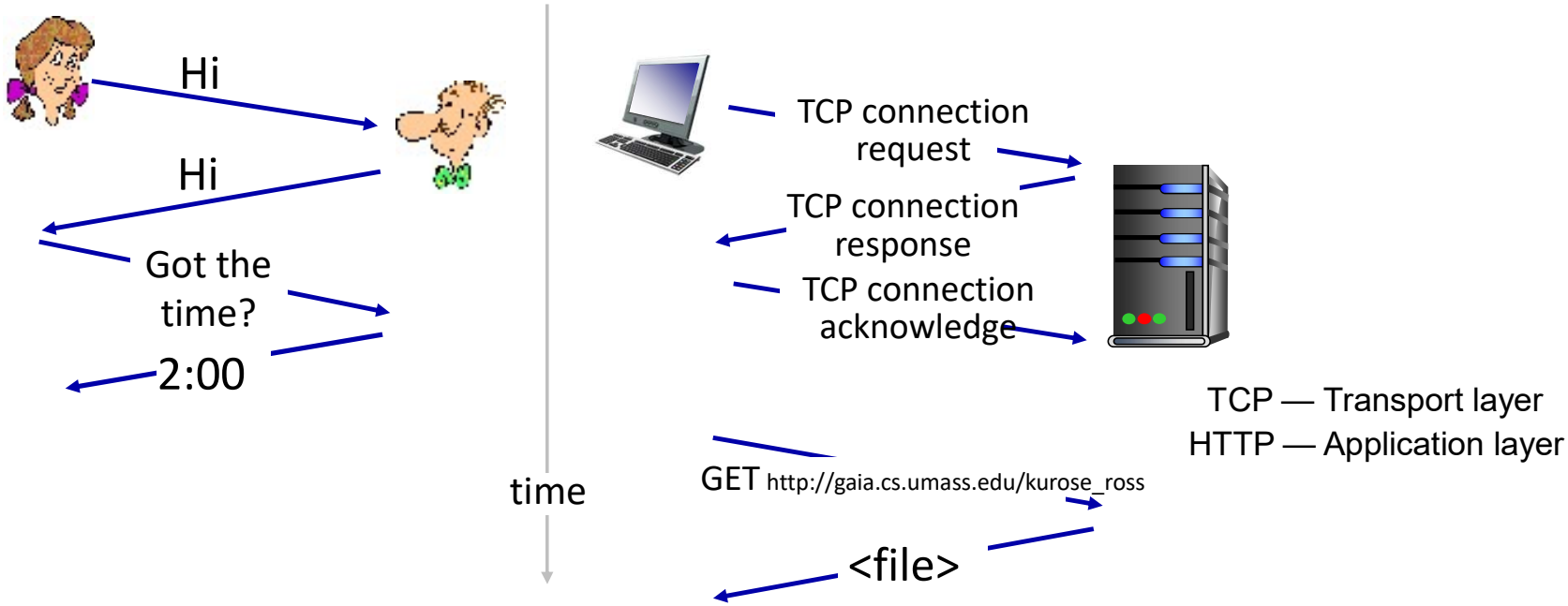
Jakobson communication model





Human and network protocols

A human protocol and a computer network protocol:



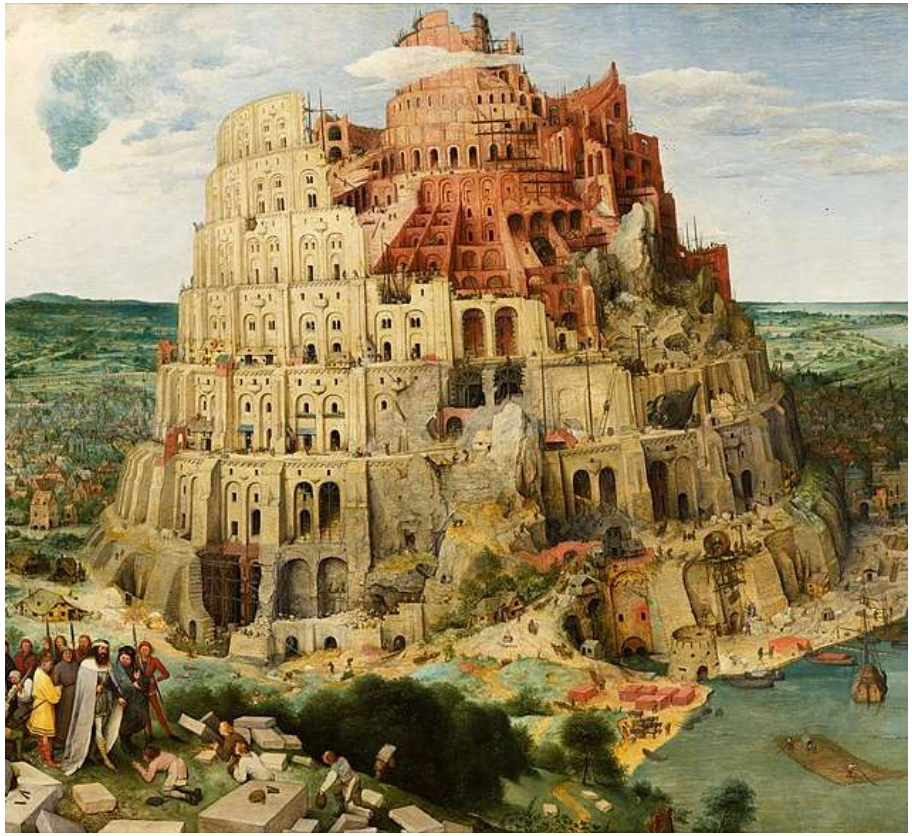
Introduction

Needs for Standards and Protocols

- Computer networks (such as Internet) can be quite **complex and widespread** over different locations, countries, users, etc.

- Clearly, a **set of rules** or a common ground should be defined so that all participants know how **to provide or to use services**.

- From the beginning of Internet, one of the major effort has been made to define **protocols** and **standards** that regulate communication.

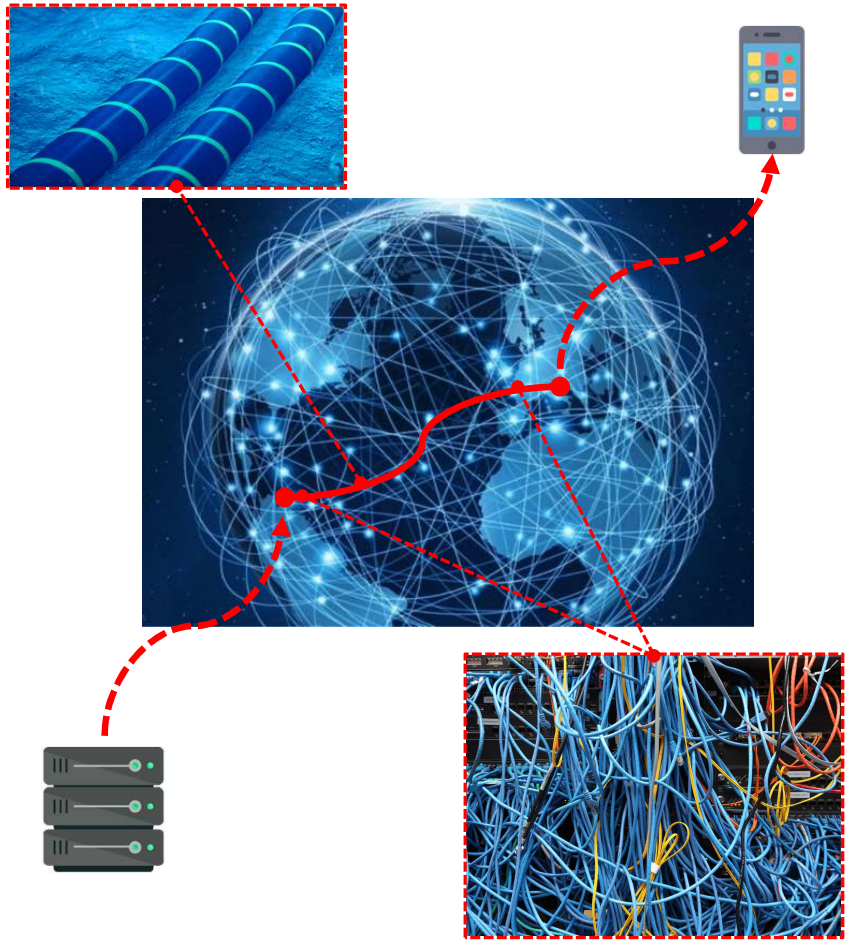


The Tower of Babel by Pieter Bruegel the Elder (1563).

Introduction

Needs for Standards and Protocols

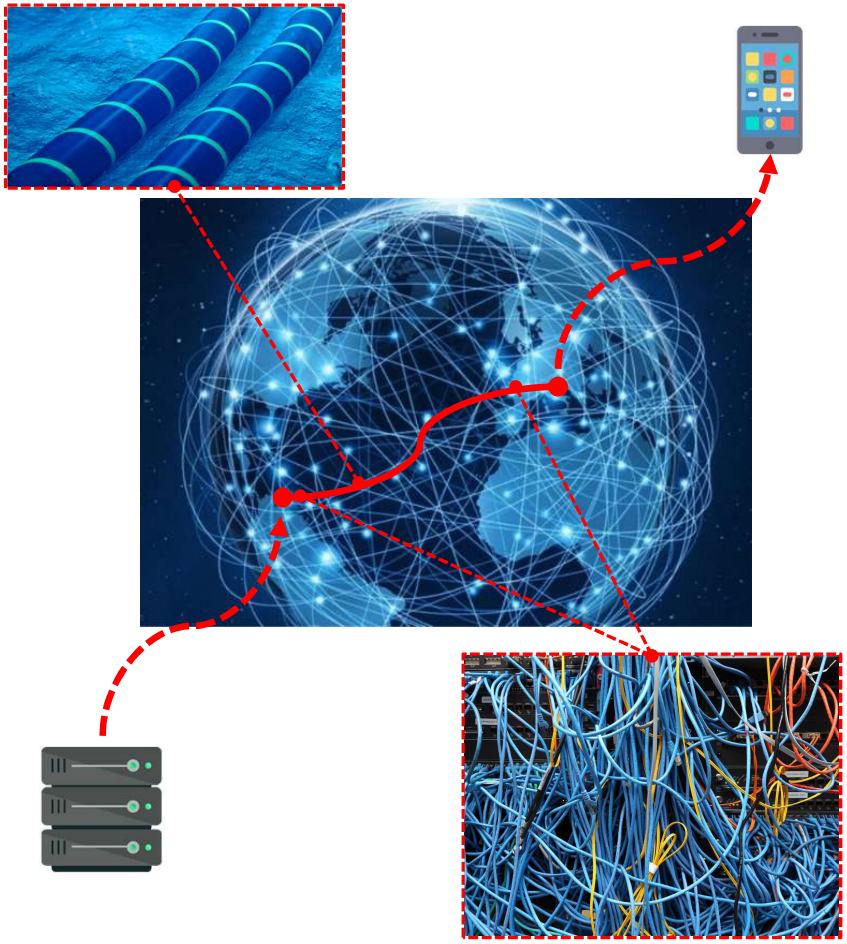
- The communication between devices involves **several problems**:
  - Physical transportation, addressing, error check, data conversion and regulation, security, synchronization, etc.
- To allow the communication both the receiver and the transmitter must agree to a **protocol**.
- When multiple/different devices from different places communicate they all must to agree on a common protocol, which becomes a **standard**.



Introduction

Needs for Standards and Protocols

- There are **hundreds of protocols** which regulate all aspects of communication from physical links to applications:
  - How **cables and links** should be created (materials, shielding, frequencies, etc.).
  - How **addresses** should be assigned.
  - How data should be wrapped into **packets or frames** for transmission.
  - How **errors** should be detected or corrected.
  - How **applications** should exchange data.
  - …



Introduction

Who Standardize Standards?

- Todays Internet standards and protocols are defined by a community of experts called **Internet Engineering Task Force** (IETF).



- IETF is typically organized in **open working groups**, each one focused on specific aspects of internet.

- Each working group produces numbered documents called **Request For Comments** (RFC) including description and definition of protocols, concepts, methods underlying a standard.
  - For example, the IPv4 protocol was defined in RFC 791 (1981).

Taming network complexity

Network architectural model

Networks are complex, with many "pieces":

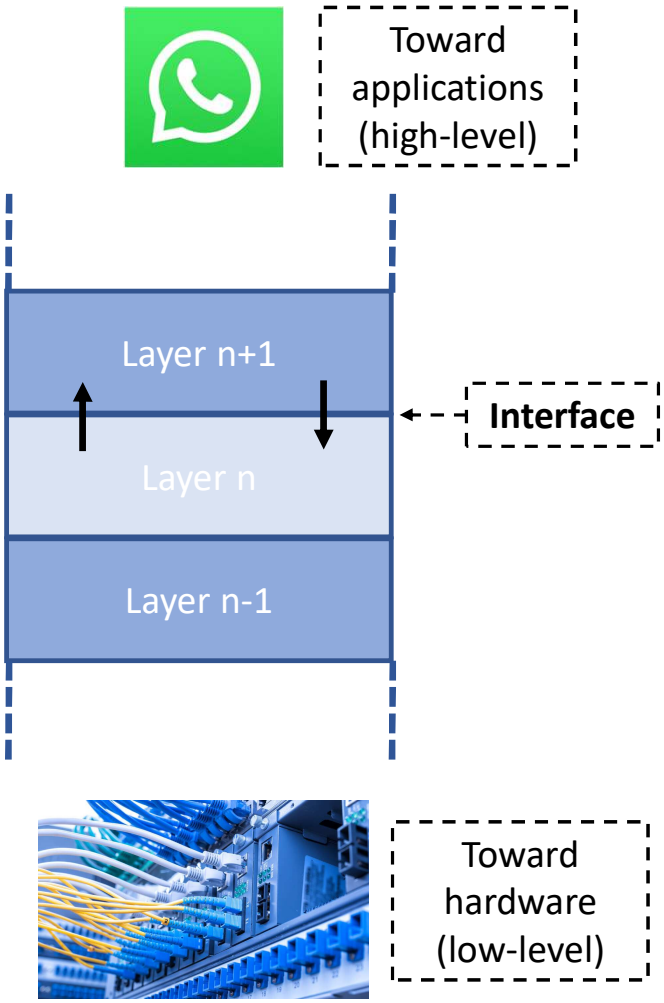
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

is there any hope of *organizing* structure of network?

Layering
Network model

- Architectural approach to implement communication: **layered model** (*divide et impera* in layers):
  - Each layer constitutes a **level of abstraction**.
  - Each layer **relies on services** from the lower layer and **provides services** to the upper layer.



Layering is not a networking idea

Network model

Dealing with complex systems:

- Operating Systems
- Information Systems
- Networks …

Layering - Discussion

Network model

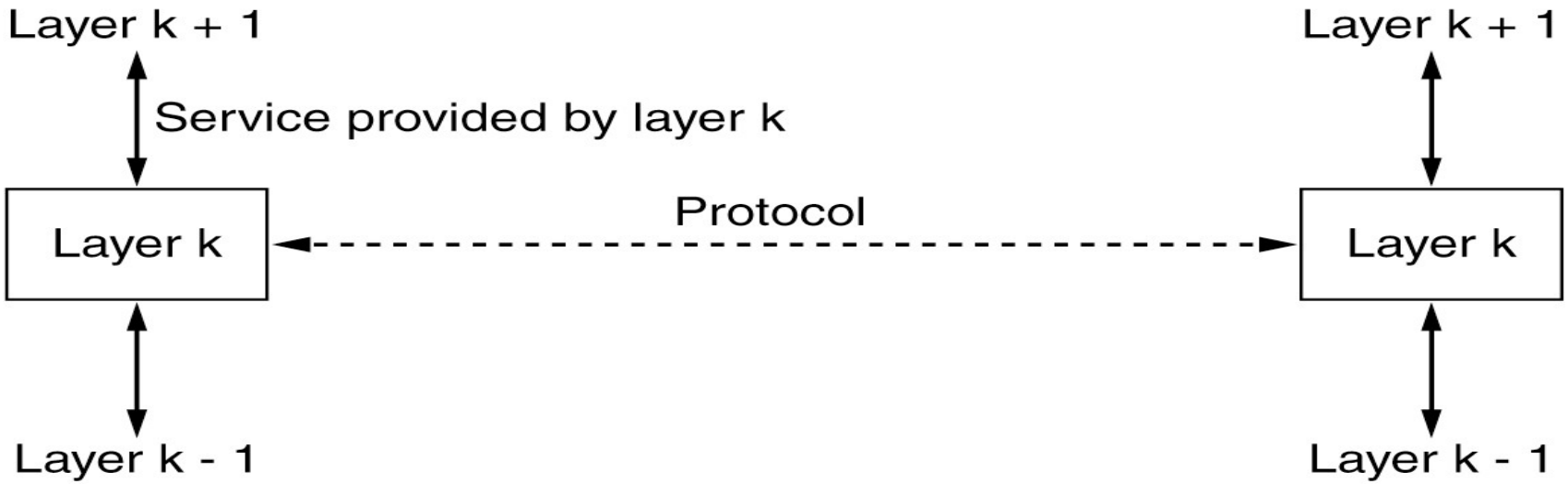
Dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change in layer's service *implementation*: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful? (efficiency)

Service primitives

Protocols and services

A **service** is set of operations a layer offers to the layer above: it says what the layer does, never how.
A **protocol** is set of rules on format and meaning of the messages exchanged with the peer entity at the same layer on the other machine.

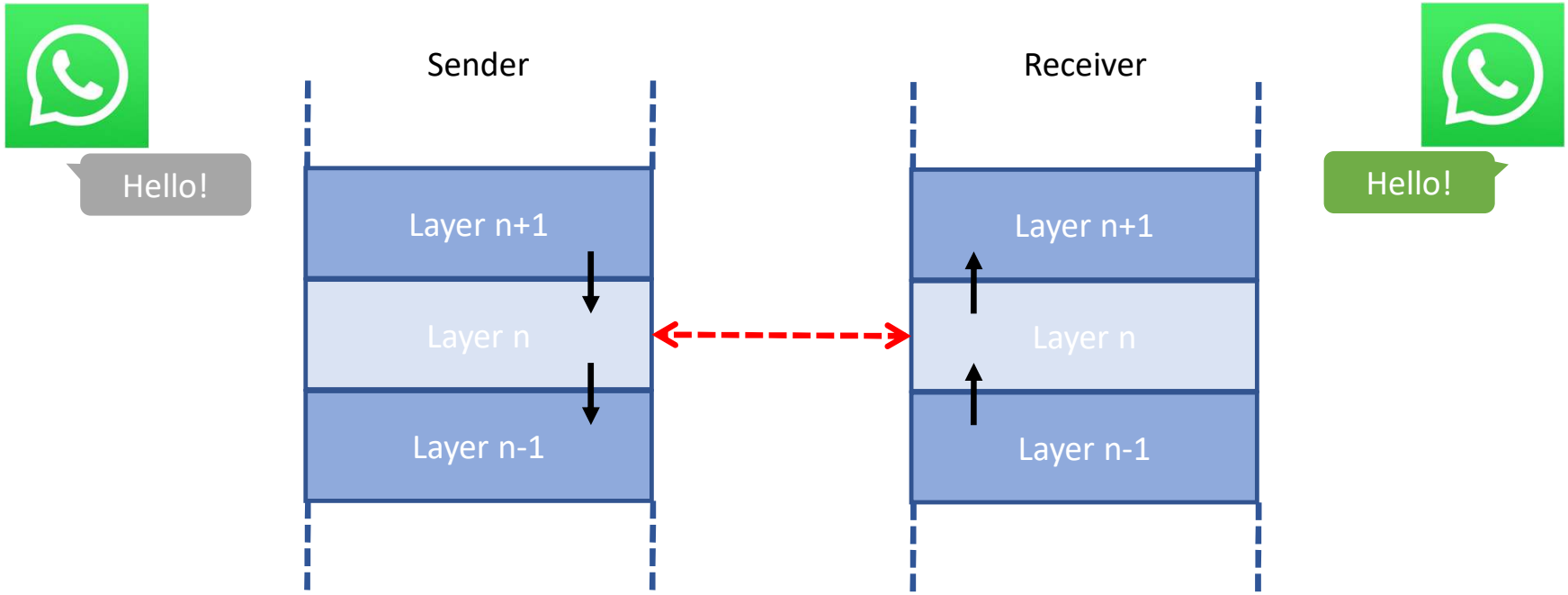


Entities use protocols in order to implement their service definitions.

Introduction

Network Models: Layered Model

- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.

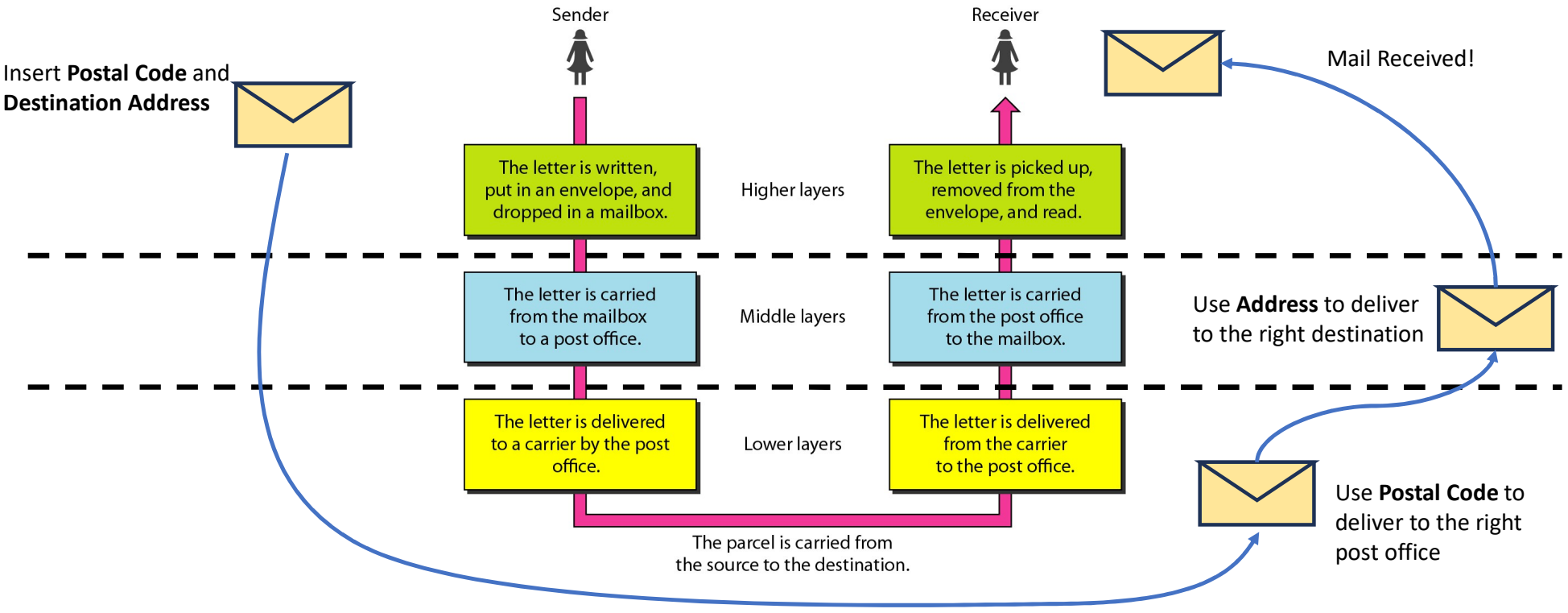


Layer *n* on the sender talks to layer *n* on the receiver, and to nothing else. That horizontal arrow is a logical conversation. Data really goes down the sender's stack and up the receiver's. Only at the physical layer does anything actually travel.

Introduction

Network Models: Layered Model (example)

- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.



Connections and reliability

- **Connection-oriented service**
  - Modeled after the **telephone system**
  - Service user first establishes a connection, uses the connection, and then releases the connection
  - The connection behaves like a tube: what goes in one end comes out the other, normally in the same order.
  - Can conduct a negotiation about the parameters to be used (maximum message size, quality of service, and so on: one side proposes, the other accepts or counters)

Connections and reliability

- **Connectionless service**
  - Modeled after the **postal system**
  - Each message carries the full destination address and is handled individually by the intermediate nodes.
  - Between the same two ends the mesh offers several paths, so two messages may take different ones — and may arrive out of order.
  - Store-and-forward switching: intermediate nodes receive a message in full before sending it on to the next node
  - Cut-through switching: transmission of a message at a node starts before it is completely received by the node
  - Datagram service: Unreliable (not acknowledged) connectionless service; named after the telegram, which also came back with no receipt.

Connections and reliability

- **Connection-oriented systems**
  - Reliable message stream:
    - message boundaries are preserved (e.g., Stream Control Transmission Protocol)
  - Reliable byte stream
    - ordered stream of bytes (e.g., file transfer / remote login)
  - Unreliable connection
    - connection is maintained, but losses may be tolerated (e.g., real-time voice/video)
- **Connectionless systems**
  - Unreliable datagram:
    - independent message, no delivery guarantee (e.g., UDP/IP-style datagram service)
  - Acknowledged datagram:
    - each independent message is acknowledged (e.g., CoAP Confirmable over UDP)

Reliability, and why you would ever give it up

Reliable service provides delivery guarantees, typically using mechanisms such as acknowledgements and retransmissions.
Both kinds of service can be reliable or not.

- Reliability mechanisms introduce overhead and may increase delay.
- Reliability may not be provided by a lower layer.
  E.g., Ethernet detects errors but does not provide ack and retransmissions to guarantee frame delivery; recovery can be performed by higher layers
- For real-time applications (e.g., VoIP), retransmitted data may arrive too late to be useful. In this case, low delay can be more important than perfect delivery.

Reliable services can be built on top of unreliable ones
E.g., TCP provides reliable transport over best-effort IP

Connections and reliability

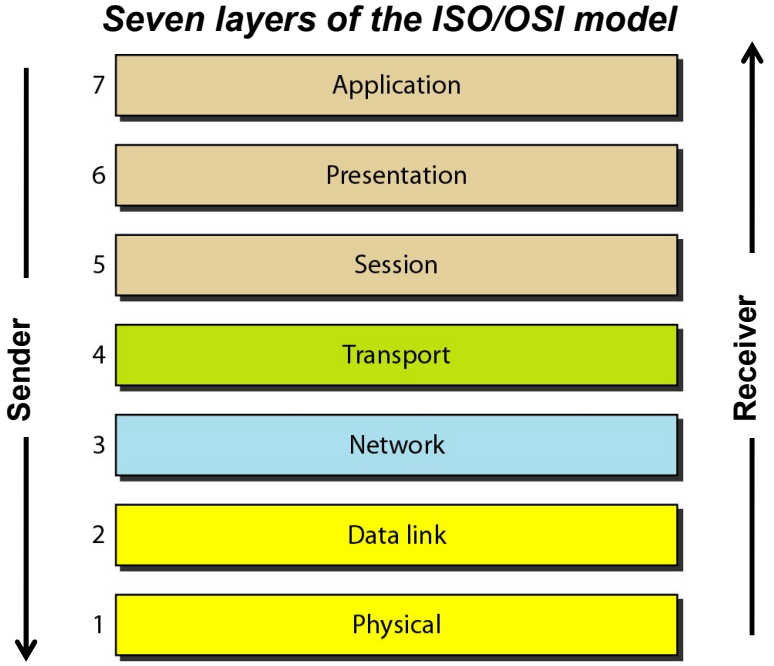
	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Six common connection-oriented and connectionless services.

Introduction

Network Models: The OSI Model

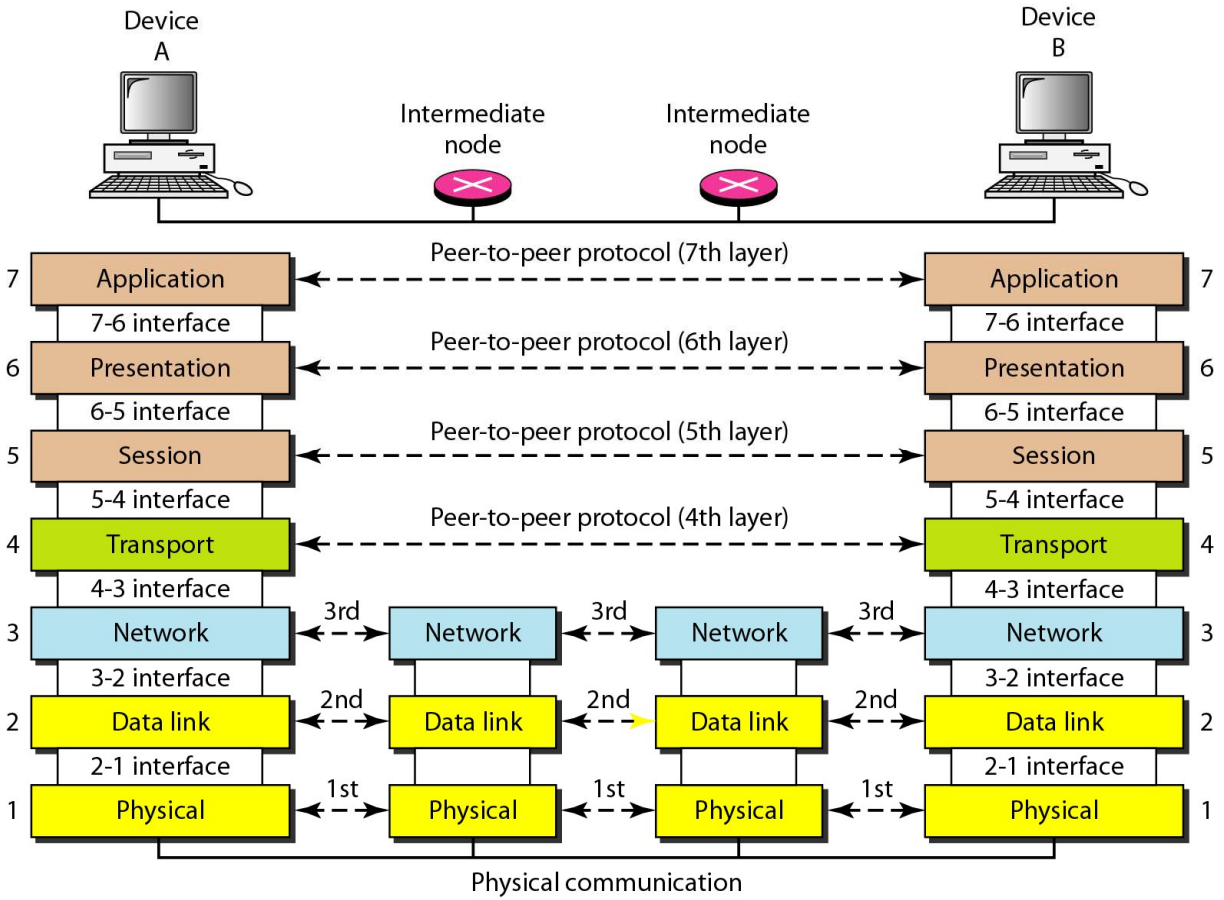
- In the ‘80 the **ISO** (International Standards Organization) defined a layered model for computer networks: the **OSI** (Open System Interconnection) model.
- The ISO/OSI model includes 7 layers:
  1. Physical transmission of raw bits.
  2. Data format definition (frames).
  3. Routing of messages through the net.
  4. Process-to-process data delivery (TCP, UDP).
  5. Session management and synchronization between communicating applications
  6. Translation of messages (encoding, compression, decryption, etc.).
  7. Data use and human-computer interaction (file sharing, emails, streaming videos, etc.).



Introduction

Network Models: The OSI Model

- Not all devices operate at all layers.
- Intermediate forwarding devices typically process only the lower 2 or 3 layers.
- End systems operate up to the Application layer.

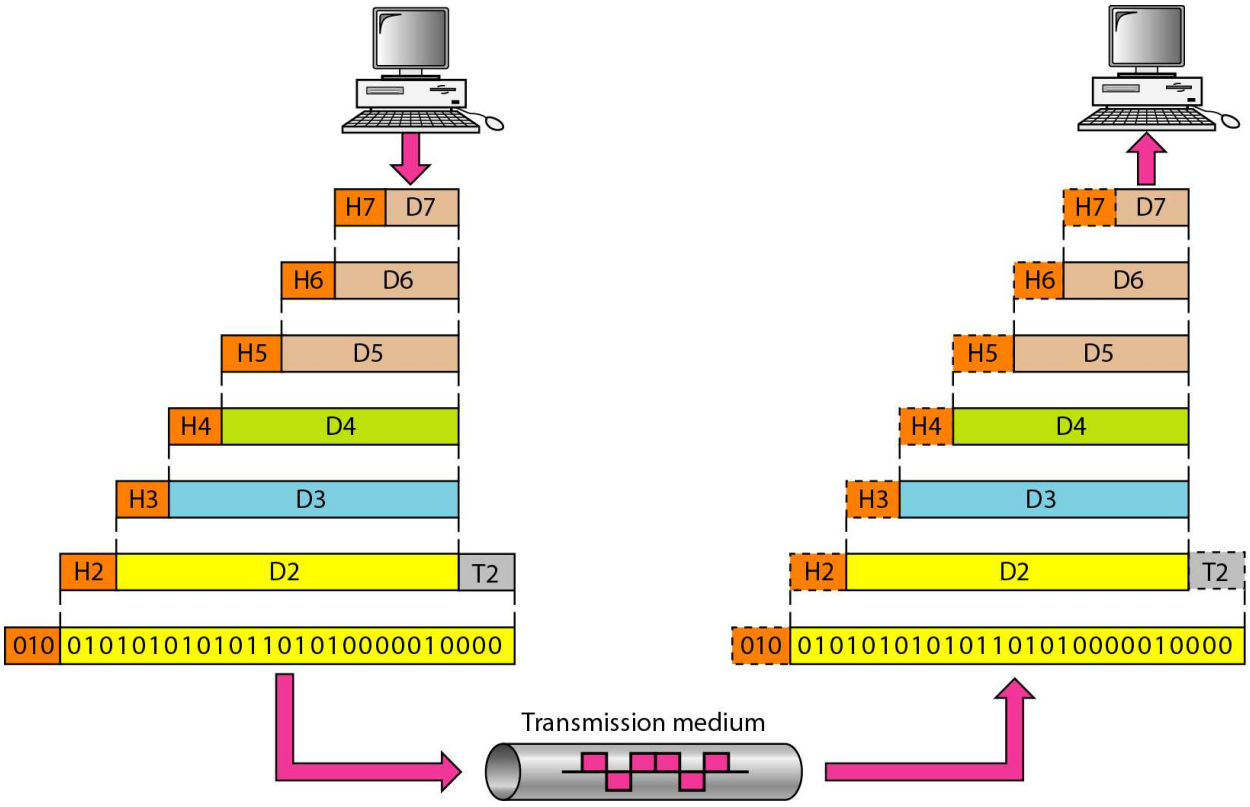


Introduction

Network Models: The OSI Model

- **Encapsulation** (top-down): every layer of the sender adds a layer-specific field to the message (payload) in the form of a header (H) or trailer (T).

- **Decapsulation** (bottom-up): Those fields are removed and interpreted by the same layer in the receiver.



Introduction

Network Models: The OSI Model

When the layer-7 message goes down, everything that came from above — header and content together — becomes the payload of layer 6.

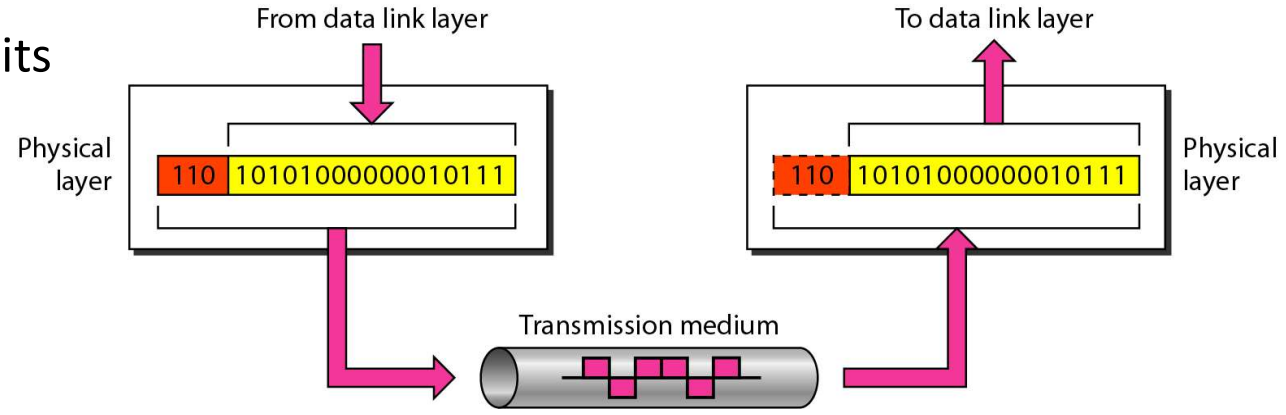
- Layer 6 prepends its own header and hands the whole thing to layer 5, which does it again.
- Each layer treats the data received from the layer above according to its own service, without depending on the internal operation of that layer.
- At the lower layers, the payload is generally treated as opaque data. For example, TCP does not need to know whether its payload contains HTTP, HTML, or something else.

The receiver walks the same sequence backwards, each layer interpreting and removing its own protocol information.

Introduction

OSI Model: 1. Physical Layer

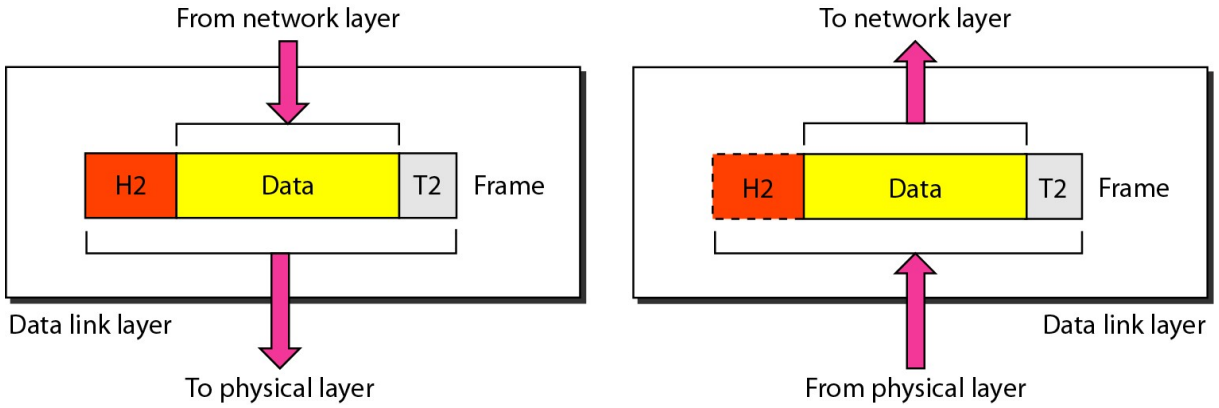
- The **physical layer** is responsible for movements of individual bits from one node to the next over a physical media
  - Physical characteristics of interfaces and media (connectors, cables, electric signals, etc.)
  - Line configuration (point-to-point or multipoint)
  - Physical topology (mesh, star, ring or bus)
  - Transmission mode (simplex, half-duplex or full-duplex)
  - Bit representation / signaling (how 0s and 1s are represented on the medium)
  - Data rate
  - Timing/synchronization of bits



Introduction

OSI Model: 2. Data Link Layer

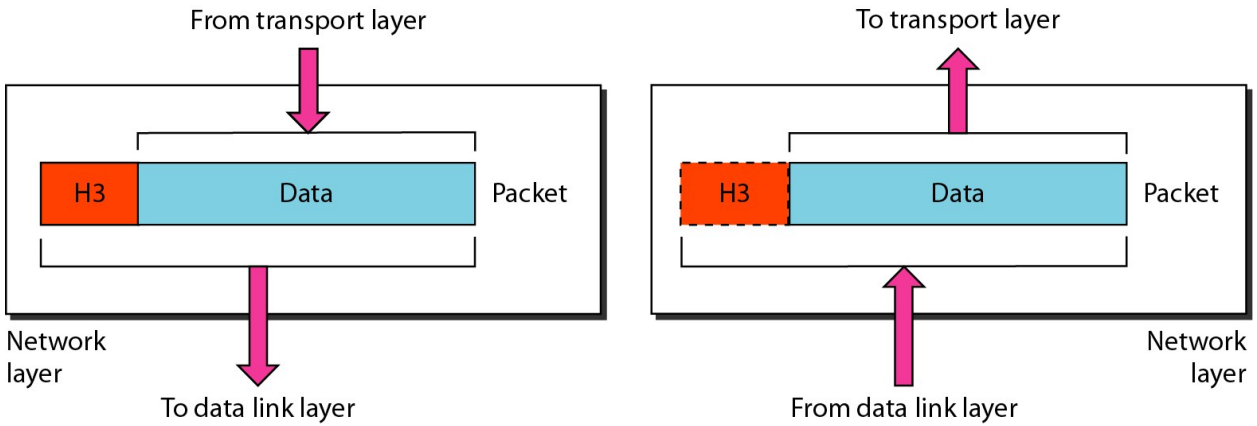
- The **data link layer** is responsible for moving frames from one node to the next node
  - **Framing:** organizes data received from the Network layer into frames
  - **Link-layer addressing** (e.g., MAC addresses)
  - **Error detection** (e.g., Frame Check Sequence, FCS) - Was the frame corrupted?
  - **Medium access control**, when the transmission medium is shared
  - **Error control** (if provided) - How do we recover from errors?
  - **Flow control** (if provided) - How do we avoid overwhelming the receiver?



Introduction

OSI Model: 3. Network Layer

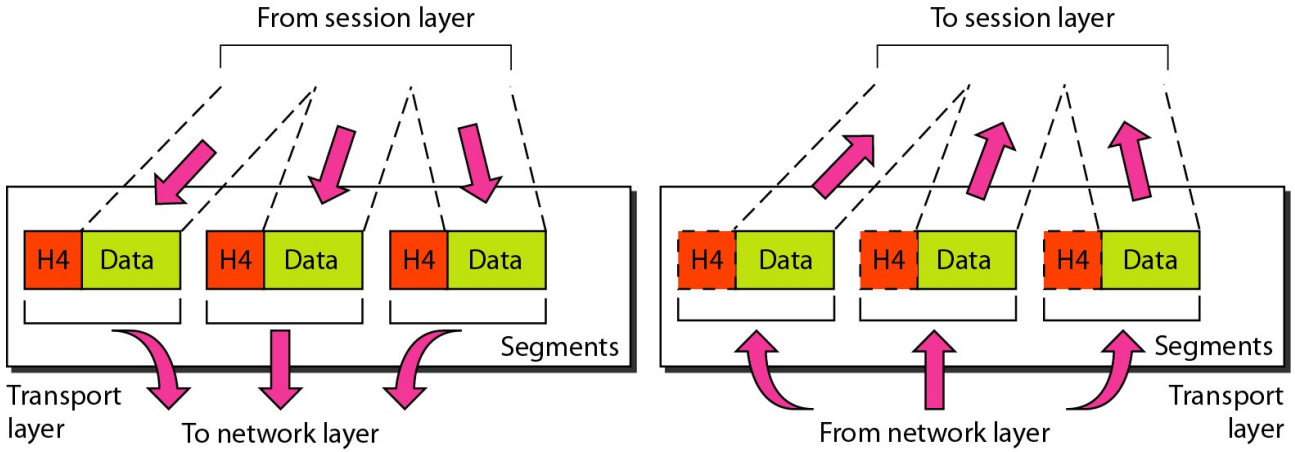
- The **network layer** is responsible for delivering packets from the source host to the destination host, possibly across multiple networks.
  - Works on **packets**
  - **Logical addressing** (e.g., IP addresses)
  - **Forwarding**: sending each packet toward the appropriate next hop
  - **Routing**: determining paths through interconnected networks



Introduction

OSI Model: 4. Transport Layer

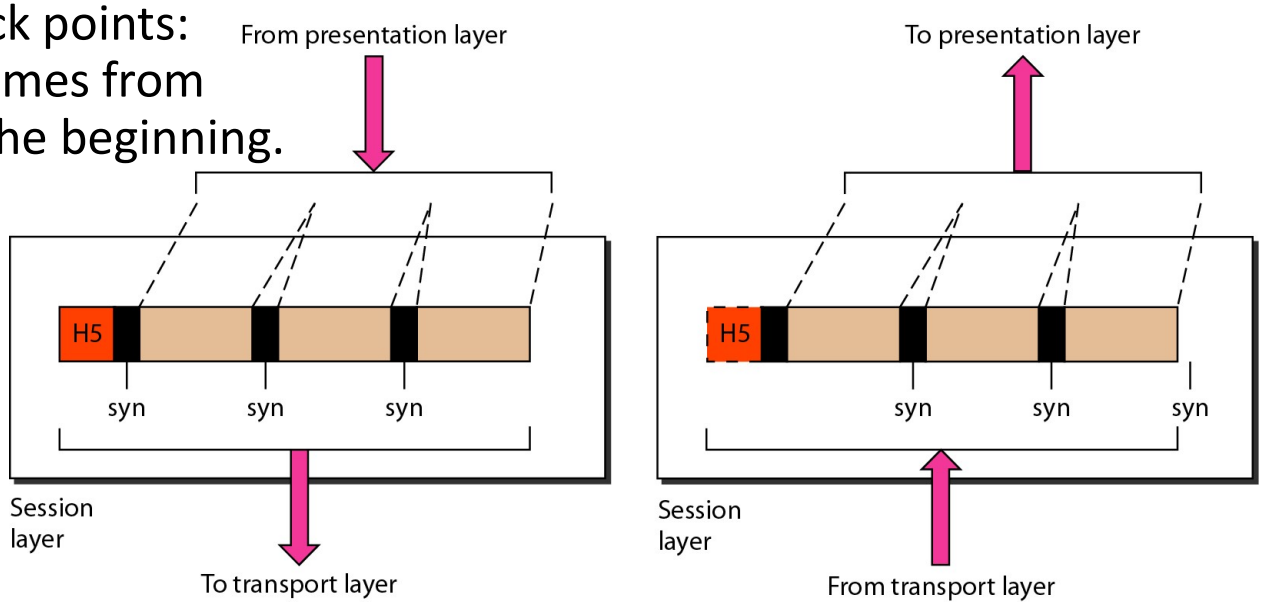
- The **transport layer** is responsible for end-to-end, process-to-process delivery of data
  - Process-to-process delivery
  - Connection control (Connection-oriented or connection-less)
  - Segmentation/reassembly
  - Port addressing (the port identifies the destination process)
  - Flow and error control, if provided



Introduction

OSI Model: 5. Session Layer

- The **Session layer** is responsible for establishing, managing, and synchronizing communication sessions between applications.
  - It establishes, maintains and synchronize the interaction between communicating system (communication session).
  - Dialog control and management
  - Synchronization through check points: an interrupted exchange resumes from the last one instead of from the beginning.

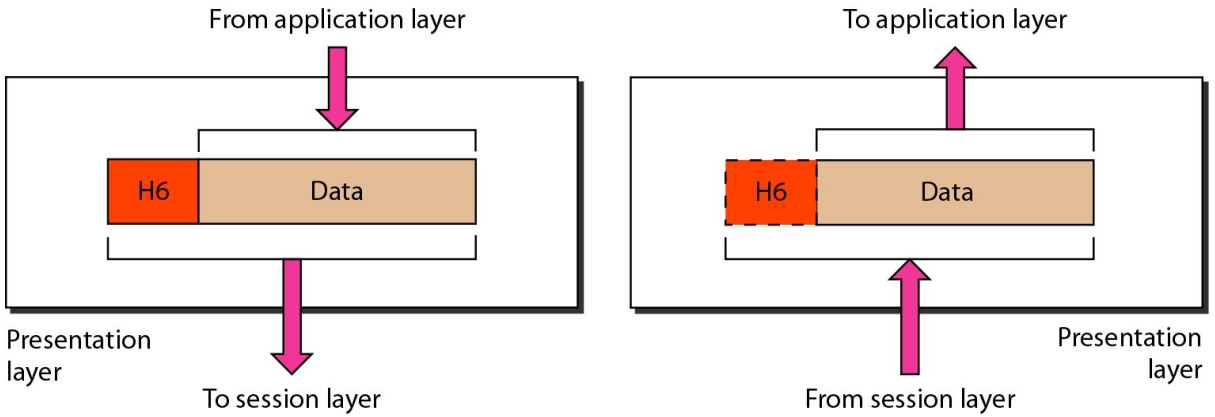


Introduction

OSI Model: 6. Presentation Layer

- The **presentation layer** is responsible for the representation of data, including translation, compression, and encryption
  - Data representation and translation (e.g., ASCII against EBCDIC)
  - Encryption and Decryption
  - Compression and decompression

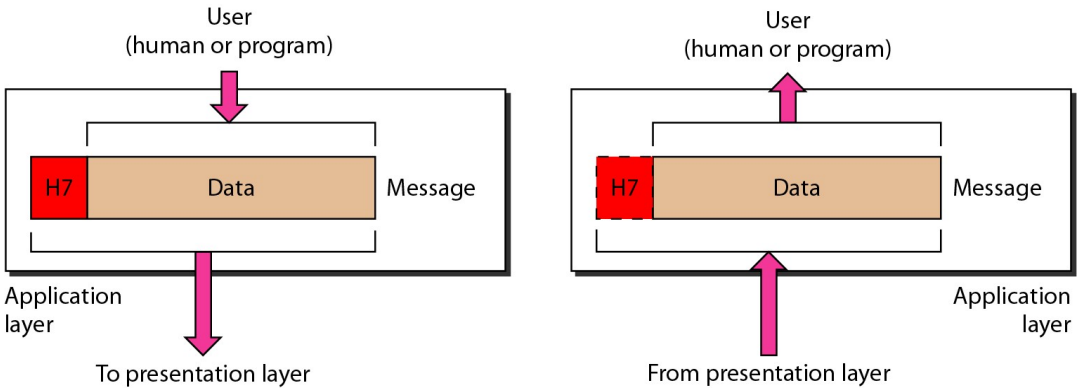
Unlike lower-layer encapsulation, Presentation-layer operations may transform the actual representation of the data, rather than simply adding a header



Introduction

OSI Model: 7. Application Layer

- The **Application layer** provides network services directly to end-user applications.
  - Web access (HTTP)
  - Electronic mail (SMTP, IMAP)
  - File transfer and access (FTP, SFTP)
  - Remote login/access (SSH)
  - Name resolution (DNS)

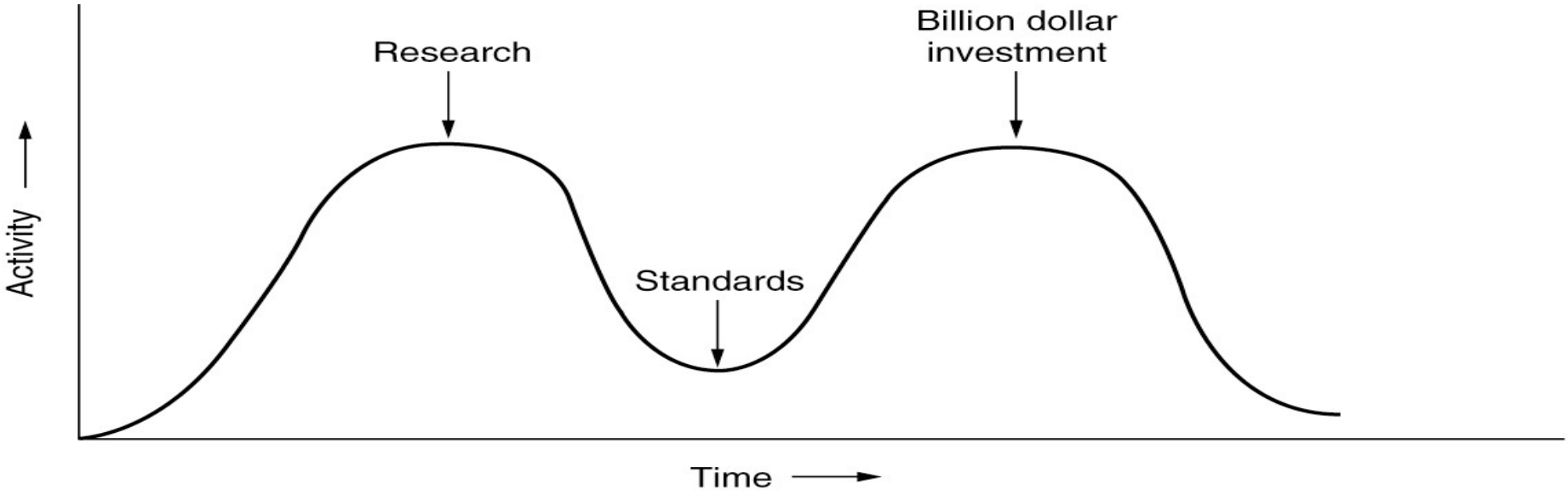


Critique of the OSI model (1/2)

- Bad timing
  - By the time OSI protocols became available, TCP/IP was already widely deployed
- Bad design
  - Some layers had overlapping or limited functionality, and the protocol suite was considered complex
- Bad implementations
  - Initial OSI implementations were often large, complex, and relatively inefficient.
- Bad politics
  - Widely thought to be the creature of the European telecommunication ministries, the European Community, and later the U.S. Government

The model survived and is still how we talk about networks. The protocols did not.

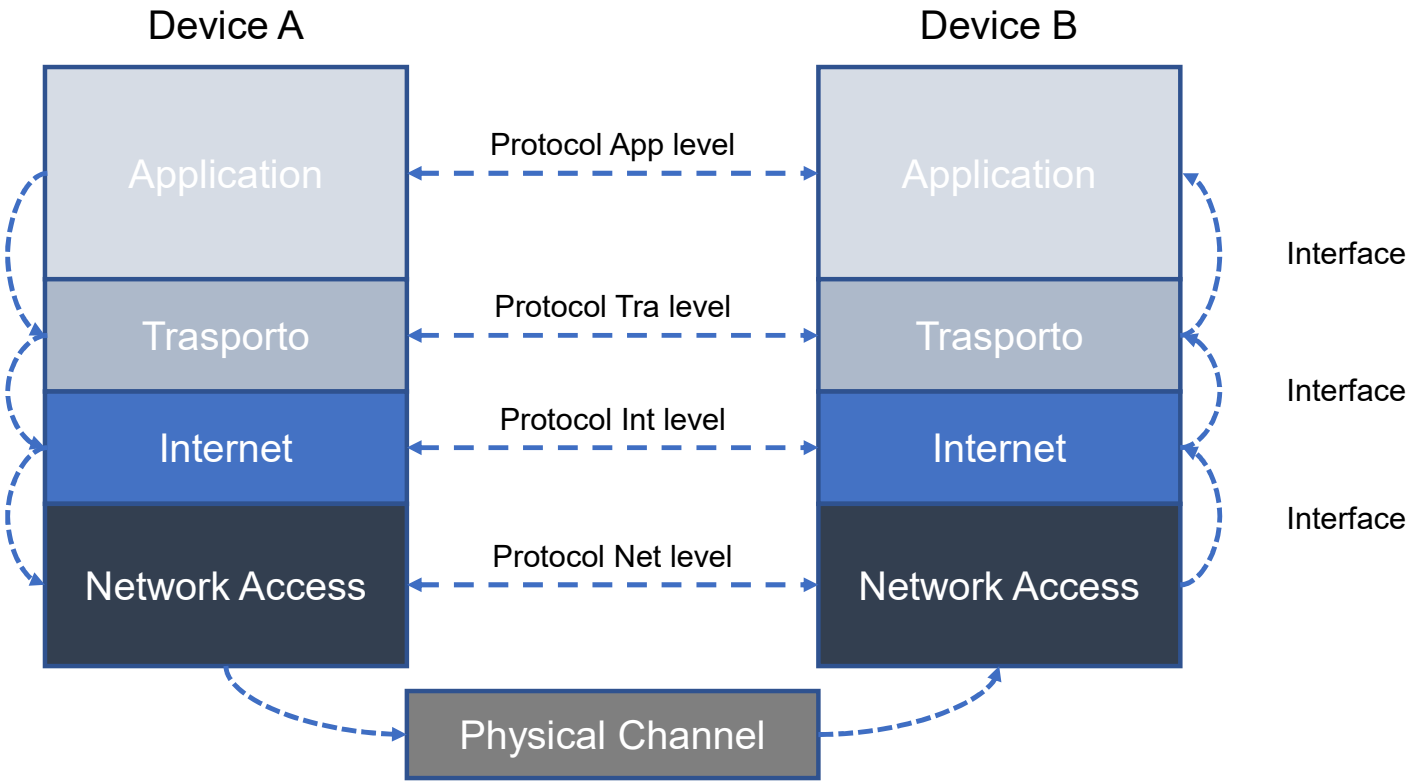
Critique of the OSI model (2/2)



To prevent bad timing, it is essential that the standards be written in the trough in between the tops of the waves known as the two ‘‘elephants."

TCP/IP Model

TCP (Transmission Control Protocol)/**IP** (Internet Protocol) model is the architecture actually used by the Internet. It is represented with four layers.



Critique of the TCP/IP model (1/2)

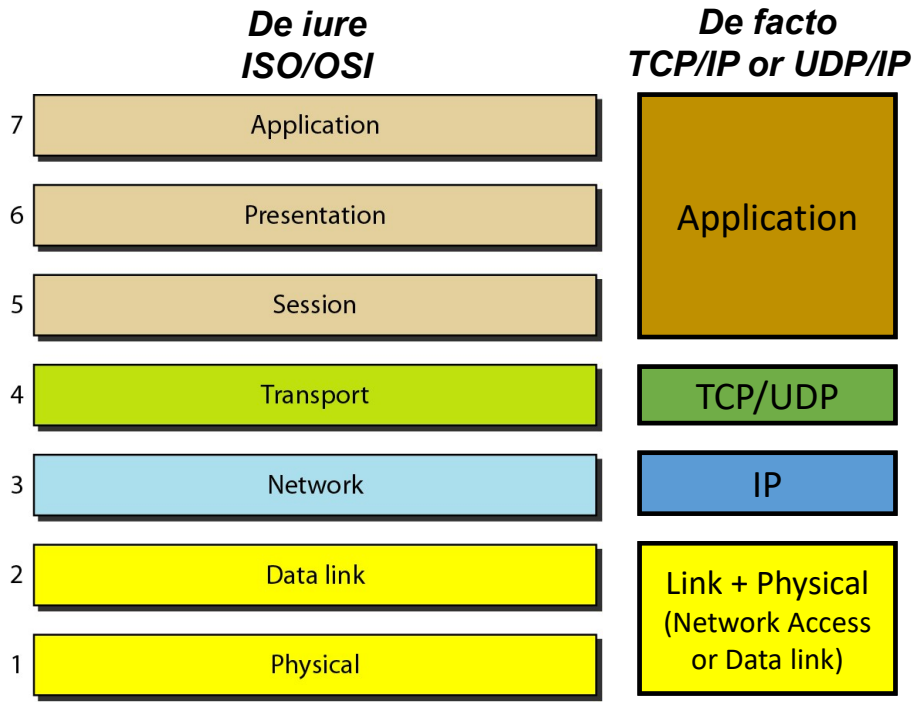
- Model does **not clearly distinguish** the concepts of services, interfaces, and protocols
- It was designed around the TCP/IP protocol suite and is **less general** than the OSI reference model.
- The **Network Access (Link) layer is only loosely spe**cified.
- The model does **not separate the physical and data link** layers

TCP/IP vs OSI

- The ISO/OSI model is a standardized reference model for network communication, but it is quite complex and detailed.

- The Internet Protocol Suite (TCP/IP) is the protocol architecture used by internet (and local networks).
  - TCP: Transmission Control Protocol.
  - UDP: User Datagram Protocol.
  - IP: Internet Protocol.

- The TCP/IP model is *de facto* the protocol architecture of the Internet



The Five-Layer Internet Model

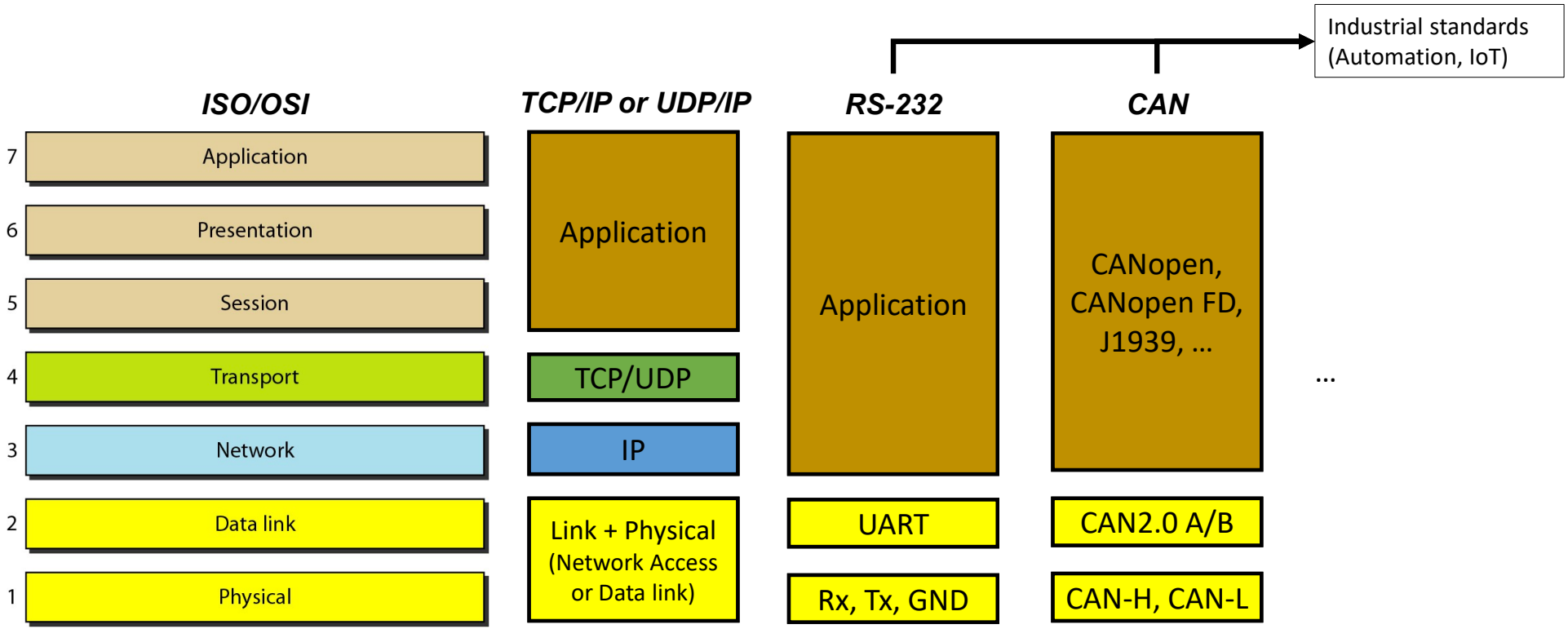
From this point on, the course uses a five-layer model: Application, Transport, Network, Link, and Physical. Session and Presentation functions are typically incorporated into the Application layer in the Internet architecture.

Layer	Corresponds to OSI	Unit of data
Application	7, 6 and 5	message
Transport	4	segment
Network	3	datagram
Link	2	frame
Physical	1	bit

We keep the OSI numbers when we need them — layer 3, layer 2 — because everyone in the field uses them. But the stack we study, top down, is this one.

Other network models

- Not all communication systems use the full Internet protocol stack. Different technologies may implement only the layers required by their application



Summary

One request crosses too many pieces to design at once, so the system is cut into layers: divide et impera.

- A layer offers a service upward through an interface, and runs a protocol sideways with its peer. The two are independent.
- Peer communication is logical: data goes down one stack and up the other, and only the physical layer moves anything.
- Encapsulation is what makes that work — each layer treats what it carries as opaque bytes.
- OSI gave us seven layers to think with, TCP/IP gave us the protocols; we use five layers and Kurose's names.

Next we open the top of the stack, where the applications are.